

Customer Order Insights & Delivery Tracker

MongoDB Component → Week 1 Deliverable

Field	Details
Name	Vaishali.S
Week	Week 1 — Database Foundation
Technology	MongoDB / mongosh
Database	supply_chain
Collection	feedback
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Week 1 Objectives

- Design MySQL tables for customers, orders, and delivery_status.
- Perform basic CRUD operations on orders.
- Write a stored procedure to fetch all delayed deliveries for a customer.
- Store customer feedback (unstructured) in MongoDB.
- Index MongoDB collection to search by customer ID.

Key Concepts Applied

6.1 insertMany()

The insertMany() method inserts multiple documents into a collection in a single operation. It is more efficient than calling insertOne() repeatedly and returns an acknowledgement object containing the auto-generated _id values.

6.2 createIndex()

Indexes in MongoDB improve the speed of query operations by reducing the number of documents MongoDB must scan. An ascending index (value: 1) on order_id allows efficient lookups and range queries on that field.

6.3 ObjectId

MongoDB automatically assigns a unique ObjectId to the _id field of each document upon insertion. ObjectIds are 12-byte identifiers that encode a timestamp, machine identifier, process ID, and a random counter.

Output

```
>_MONGOSH
> db.feedback.insertMany([
  {
    customer_id:1,
    feedback:"delivery was fast"
  },
  {
    customer_id:2,
    feedback:"late delivery and packaging issue"
  }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a0554ddccdd47dc23108a54'),
    '1': ObjectId('6a0554ddccdd47dc23108a55')
  }
}
> db.feedback.createIndex({customer_id:1})
< customer_id_1
> db.feedback.find()
< {
  _id: ObjectId('6a0554ddccdd47dc23108a54'),
  customer_id: 1,
  feedback: 'delivery was fast'
}
{
  _id: ObjectId('6a0554ddccdd47dc23108a55'),
  customer_id: 2,
  feedback: 'late delivery and packaging issue'
```